

Engineering Mathematics 1



STD Number – IT26101833

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Lab 01;

Exercise 3:

Exercise3.mExercise1.m

```
1 close; %close all the open graphs
2 clear; %clear all the variables
3 clc; %clear the command window
4
5 a=2+2i; b=4-5i;
6 a_x= real(a); a_y=imag(a);
7 b_x=real(b); b_y=imag(b);
8
9 %printing the real and imaginary part of the above assigned complex numbers
10
11 fprintf("a_x = %f\n", a_x);
12 fprintf("a_y = %f\n", a_y);
13 fprintf("b_x = %f\n", b_x);
14 fprintf("b_y = %f\n", b_y);
15
16 printf("Modulus of a=%2f\n", abs(a)); %modulus of the complex number a
17 printf("Argument of a=%2f degrees\n", (180/pi)*arg(a)); %argument of the complex number a
18 printf("Modulus of b=%2f\n", abs(b)); %modulus of the complex number b
19 printf("Argument of b=%2f degrees \n", (180/pi)*arg(b)); %argument of the complex number b
20
21 figure
22 plot([0 a_x], [0 a_y], [0 b_x], [0 b_y]); % calling the plot function
23 set(gca, "xaxislocation", "origin", "yaxislocation", "origin"); % set the axes scale to go through (0,0).
24 xlim([-10 20]); ylim([-10 10]); %setting the x and y axes limits.
25 title('Argaon Diagram'); %set the title of the diagram
26 xlabel('\itReal'); ylabel('\itImaginary'); %naming the axes
27 text(a_x, a_y, ["(" num2str(real(a)) ", " num2str(imag(a)) ")"]); % create a text label for the coordinate of the complex nnumbers
28 text(b_x, b_y, ["(" num2str(real(b)) ", " num2str(imag(b)) ")"]);
29
30
```

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```
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27 text(a_x, a_y, ["(" num2str(real(a)) ", " num2str(imag(a)) ")"]); % create a text label for the coordinate of the complex nnumbers
28 text(b_x, b_y, ["(" num2str(real(b)) ", " num2str(imag(b)) ")"]);
29 grid('on'); legend('a', 'b'); % turn on the grid lines
30
```

line: 30 col: 1 encoding: SYSTEM eol: CRLF

Command Window Documentation Editor Variable Editor

Activate Windows
Go to Settings to activate Windows.

```

Command Window
a_x = 2.000000
a_y = 2.000000
b_x = 4.000000
b_y = -5.000000
Modulus of a=2.83
Argument of a=45.00 degrees
Modulus of b=6.40
Argument of b=-51.34 degrees
>> |

```

